

Frequency of anthropometric implausible values estimated from different methodologies: a systematic review and meta-analysis

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Context: Poor anthropometric data quality affect the prevalence of malnutrition and could harm public policy planning. **Objective:** This systematic review and meta-analysis was designed to identify different methods to evaluate and clean anthropometric data, and to calculate the frequency of implausible values for weight and height obtained from these methodologies. **Data Sources:** Studies about anthropometric data quality and/or anthropometric data cleaning were searched for in the MEDLINE, LILACS, SciELO, Embase, Scopus, Web of Science, and Google Scholar databases in October 2020 and updated in January 2023. In addition, references of included studies were searched for the identification of potentially eligible studies. **Data Extraction:** Paired researchers selected studies, extracted data, and critically appraised the selected publications. **Data Analysis:** Meta-analysis of the frequency of implausible values and 95% confidence interval (CI) was estimated. Heterogeneity (I^2) and publication bias were examined by meta-regression and funnel plot, respectively. **Results:** In the qualitative synthesis, 123 reports from 104 studies were included, and in the quantitative synthesis, 23 studies of weight and 14 studies of height were included. The study reports were published between 1980 and 2022. The frequency of implausible values for weight was 0.55% (95%CI, 0.29–0.91) and for height was 1.20% (95%CI, 0.44–2.33). Heterogeneity was not affected by the methodological quality score of the studies and publication bias was discarded. **Conclusions:** Height had twice the frequency of implausible values compared with weight. Using a set of indicators of quality to evaluate anthropometric data is better than using indicators singly.

Systematic Review Registration: PROSPERO registration no. CRD42020208977.

Key words: anthropometry, biologically implausible values, data cleaning, outlier, quality.

INTRODUCTION

An increase in anthropometric data available from electronic health records systems was recently seen mainly in weight and height variables.¹ This large amount of information has imposed on investigators the challenge of

dealing with data quality. Anthropometric data must be reliable so the prevalence of malnutrition conditions will represent the current nutritional status of the population² and the information can be used in the public policy process.³ Measurement error is a challenging situation to keep the data reliable; however, it will be necessary to deal with

it in any method that targets to measure a determined event. In anthropometrics, there are 4 recognized possible causes of measurement error: (1) the body part to be measured, (2) the equipment used, (3) the measurement procedures, and (4) the professional performance.⁴ During data collection in surveys, researchers follow standardized protocols to avoid systematic errors and minimize random ones.⁵ There is evidence, however, of poor anthropometric data quality in some population surveys.⁶

To manage this challenge, the World Health Organization (WHO) recommends analyzing the following set of indicators for data quality assessment in surveys, which can be adapted for electronic health records systems: completeness, sex ratio, age heaping, digit preference of measurements, implausible *z*-score values, standard deviation (SD) of *z* scores, and normality (skewness and kurtosis) of *z* scores.⁷ Beyond understanding the level of data quality, methodologies to clean anthropometric data allow the nutritional status of the population to be properly characterized. Methods to clean anthropometric data rely on the logic of removing values outside of a determined limit.⁸ They could be organized into 3 groups: (1) externally defined standards based on normative references or on strict cutoffs, (2) internally defined standards based on sample-specific criteria, and (3) a combination of both (external and internal standards).⁸ In that way, cross-sectional designs allow using only methods to identify implausible values based on population references or on strict cutoffs, whereas longitudinal designs allow the use of both population references or strict cutoffs and individual trajectories.⁸

A consensus on the operational issue for data cleaning is far from being reached among researchers because it is questionable what the limit is at which residual variability is a function of flagging methodology. There is some evidence that the method chosen for cleaning anthropometric data may influence the prevalence of nutritional diagnoses.^{9–11}

In the context of demographic and nutritional transitions and the data analysis of secular trends of nutritional indicators, anthropometric data reliability is key for monitoring nutritional status. Therefore, the present systematic review and meta-analysis were designed to identify the different methods to evaluate and to clean anthropometric data, and to calculate the frequency of implausible values for weight and height obtained from these methodologies.

MATERIAL AND METHODS

Protocol and registration

The protocol with the methods of this systematic review was registered *a priori* in the International Prospective

Register of Systematic Reviews (PROSPERO; registration no. CRD42020208977). This review followed the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines.¹²

Eligibility criteria

In this systematic review, we evaluated original, observational, and interventional studies in humans of all age groups that involved the issues of anthropometric data quality and/or anthropometric data cleaning. In this study, anthropometric data quality concerns any methodology used to understand and/or discriminate the reliability level of the anthropometric data. Anthropometric data cleaning, in turn, refers to any methodology used to signal and remove anthropometric data considered implausible. The participants, intervention, comparison, outcome, and study design (PICOS) information is presented in [Table 1](#).

Information sources

The MEDLINE, LILACS, SciELO, Embase, Scopus, Web of Science, and Google Scholar databases were searched. There were no restrictions according to the date of publication, but only articles published English, French, Portuguese, and Spanish were included. In addition, references of included studies were searched for the identification of potentially eligible studies.

Search strategy

A search strategy was planned and applied in October 2020 and updated in January 2023. The terms used for each database in the systematic searches are presented in [Appendix S1 in the Supporting Information online](#). The search results were organized in an Excel spreadsheet (Microsoft) for removing duplications and for subsequent steps of the systematic review.

Study selection

Screening of titles and abstracts for eligibility was conducted by 2 researchers independently. Afterward, paired researchers independently performed the full-text assessment. Disagreements were resolved by consensus after discussion.

Data collection process

Data were extracted to a standardized Excel spreadsheet independently by paired researchers. Data on the following items were collected: title, place, authors, year of publication, objective, year of data collection, design,

Table 1 PICOS criteria for inclusion of studies

Parameter	Description
Participants	All age groups
Intervention	Not applicable
Comparison	Not applicable
Outcome	Frequency of implausible values and/or anthropometric data quality analysis
Study design	No restrictions: observational studies, interventional studies

age group, sex, race/ethnicity, sample size, data quality assessment method, a summary of measurement quality indicators, and frequency of implausible values. Disagreements were resolved by consensus after discussion.

Risk of bias in individual studies

The risk of bias was evaluated by paired researchers independently, using the scale of Loney et al.¹³ This scale was chosen because it was designed for both prevalence and incidence studies, and consisted of the following items: (a) methods validity (n = 6 items), (2) interpretation of the results (n = 1 item), and (3) applicability of the results (n = 1 item). The item about the adequacy of sample size was adapted to actualize the scale to the current use of big databases. Disagreements were resolved by consensus after discussion. The complete scale is presented in [Appendix S2 in the Supporting Information online](#).

Summary measures

The primary outcome was the frequency of implausible values for weight and height and its 95% confidence intervals. Secondary outcomes included the frequency of implausible values for weight and height according to study design, implausible values identification method, and data collection source.

Synthesis of results

The meta-analysis of proportions was conducted with Freeman–Tukey double arcsine transformation,¹⁴ using the metaprop command in Stata (StataCorp).¹⁵ The I^2 index was estimated to evaluate heterogeneity among studies. Publication bias was evaluated by funnel plot asymmetry.¹⁶ Meta-regression was performed to analyze the effect of quality score on the variability of the frequency of implausible values among studies. Statistical analyses were performed using Stata SE, version 17.0 (StataCorp).

RESULTS

Methods to evaluate anthropometric data quality

Of 6927 retrieved records, the full text of 131 were assessed, and 123 reports from 104 studies were included in the qualitative analysis. In the quantitative synthesis, 23 studies on weight^{1,10,17–37} and 14 studies on height^{1,10,19,20,23,24,26,27,30,31,34,38–40} were included ([Figure 1](#)). The references of reports of included studies are available in [Appendix S3 in the Supporting Information online](#). Listed in [Appendix S4 in the Supporting Information online](#) are the references of reports assessed for data extraction excluded and their respective reason for exclusion.

[Table S1 in the Supporting Information online](#) lists the individual characteristics of each included report (author and year, study design, age group, sex, sample size, anthropometric data quality assessment method, and cutoff used for the identification of implausible values). The year of publication of the reports varied from 1980 to 2022, and cross-sectional and longitudinal studies were the most frequent. The following quality indicators were identified: frequency of implausible values; completeness; digit preference of height and weight; mean, standard deviation, kurtosis, and skewness of z scores; age and sex ratio; Bland–Altman plot; mean absolute differences; intraclass correlation coefficient; heaping of age; intra- and interobserver error; and correction of negative values and transposition error.

In [Table S1 in the Supporting Information online](#), the methods that used population approaches or absolute cutoffs are categorized as cross-sectional, and the methods that used individual approaches are categorized as longitudinal. The data cleaning methods can be distinguished in the following resulting groups: (1) cross-sectional with internal reference; (2) cross-sectional with external reference (namely, National Center for Health Statistics growth chart; Centers for Disease Control and Prevention 2000 growth chart; Child Growth Foundation 1997 body mass index; WHO 2006 child growth standards; WHO 2007 growth reference; Health Survey for England; and WHO 2006 child growth standards with different flags [ie, Standardized Monitoring and Assessment of Relief and Transitions (SMART), WHO fixed and flexible, and EPI Info]; (3) cross-sectional with absolute cutoff; (4) longitudinal with different approaches to model data; and (5) use of algorithms based, or not, in artificial intelligence.

Critical appraisal

Studies were limited mainly in the full description of the methodology used to evaluate anthropometric data

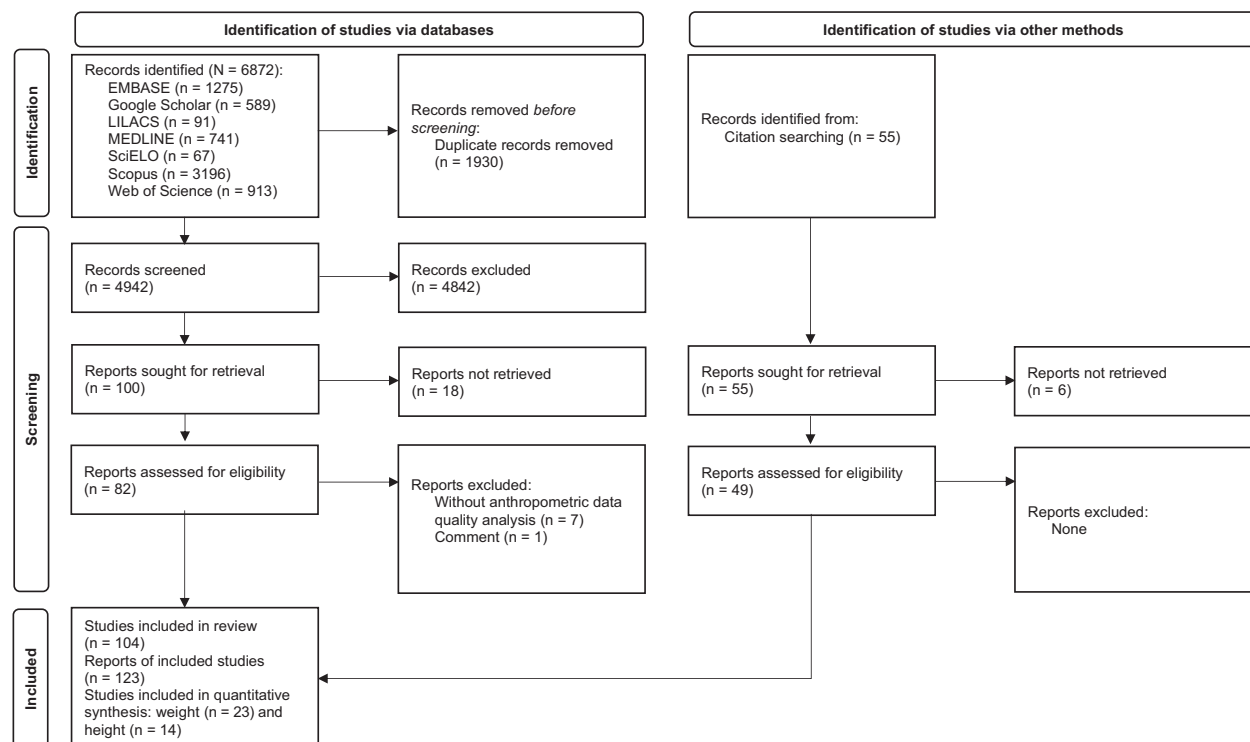


Figure 1 Flow diagram of the literature search process.

quality and to clean the data if necessary. The quality assessment score ranged from 1 to 9 with a median of 6. The critical appraisal for each report is available in [Table S2 in the Supporting Information online](#).

Quantitative synthesis: implausible values for weight

The frequency of implausible values for weight was 0.55% (95% confidence interval, 0.29–0.91; $I^2 = 99.99\%$) ([Figure 2](#)).^{1,10,17–37} The frequency of implausible values was similar between study designs (see [Figure S1 in the Supporting Information online](#)) and among different approaches to sign implausible values (see [Figure S2 in the Supporting Information online](#)), but different between data collection sources: the frequency of implausible values was lower in surveys (0.22%) than in electronic health records systems (0.70%) ([Figure 3](#)).^{1,10,17–37} The frequency of implausible values also was lower in adults than in children and adolescents (see [Figure S3 in the Supporting Information online](#)). The variability in the frequency of implausible values was not explained to a statistically significant level by the methodological quality score of the studies ($P = 0.702$; residual $I^2 = 99.02\%$) ([Figure 4](#)). A symmetric distribution was observed in the funnel plot without evidence of small studies' effect on the frequency of implausible values for the weight (see [Figure S4 in the Supporting Information online](#)).

Quantitative synthesis: implausible values for height

The frequency of implausible values for height was 1.20% (95% confidence interval, 0.44%–2.33%; $I^2 = 100.00\%$) ([Figure 5](#)).^{1,10,19,20,23,24,26,27,30,31,34,38–40} The frequency of implausible values was similar between study designs (see [Figure S5 in the Supporting Information online](#)) and apparently greater in children than in adults (see [Figure S6 in the Supporting Information online](#)). The variability in the frequency of implausible values was not explained to a statistically significant level by the methodological quality score of the studies ($P = 0.927$; residual $I^2 = 99.53\%$) ([Figure 6](#)). A symmetric distribution was observed in the funnel plot without evidence of small studies' effect on the frequency of implausible values for height (see [Figure S7 in the Supporting Information online](#)).

DISCUSSION

In this study, the literature about methods to evaluate and to clean anthropometric data was reviewed, and the frequency of implausible values for weight and height was analyzed. Some issues deserve to be highlighted. First, a set of indicators to characterize anthropometric data was observed, as were attempts to summarize them. Second, cross-sectional methods to clean anthropometric data were used most in the reports, and robust cleaning methods were mainly used in longitudinal

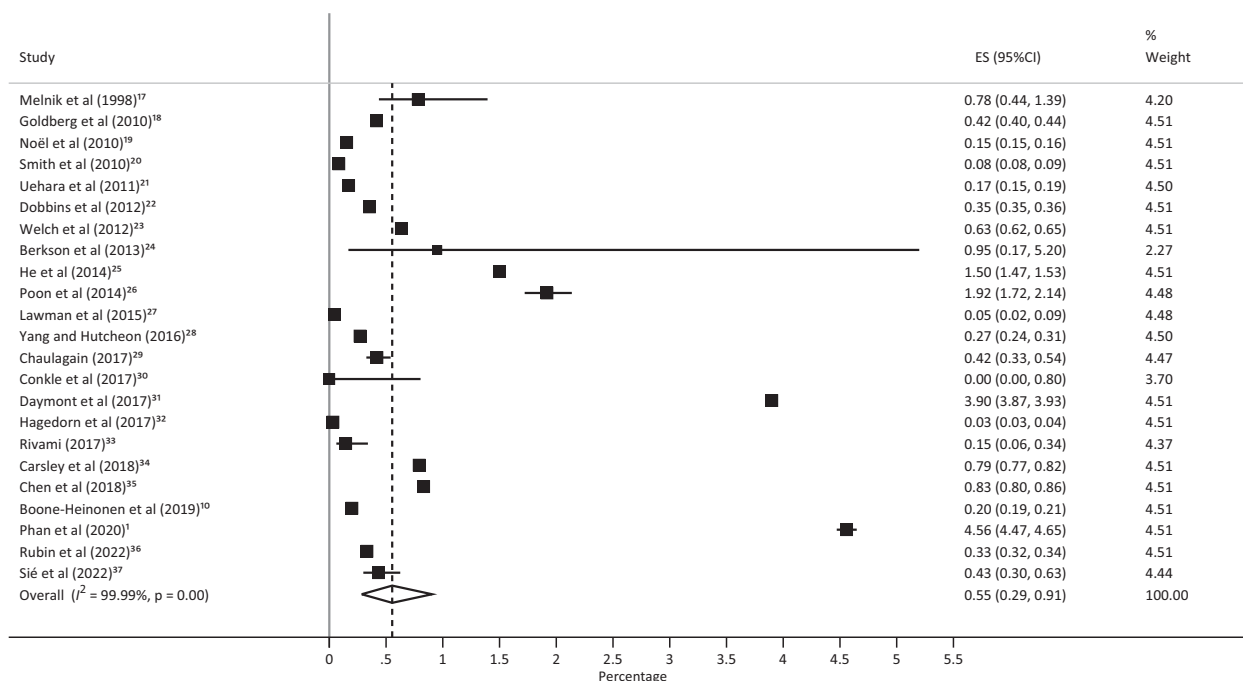


Figure 2 Frequency of implausible values for weight measure.

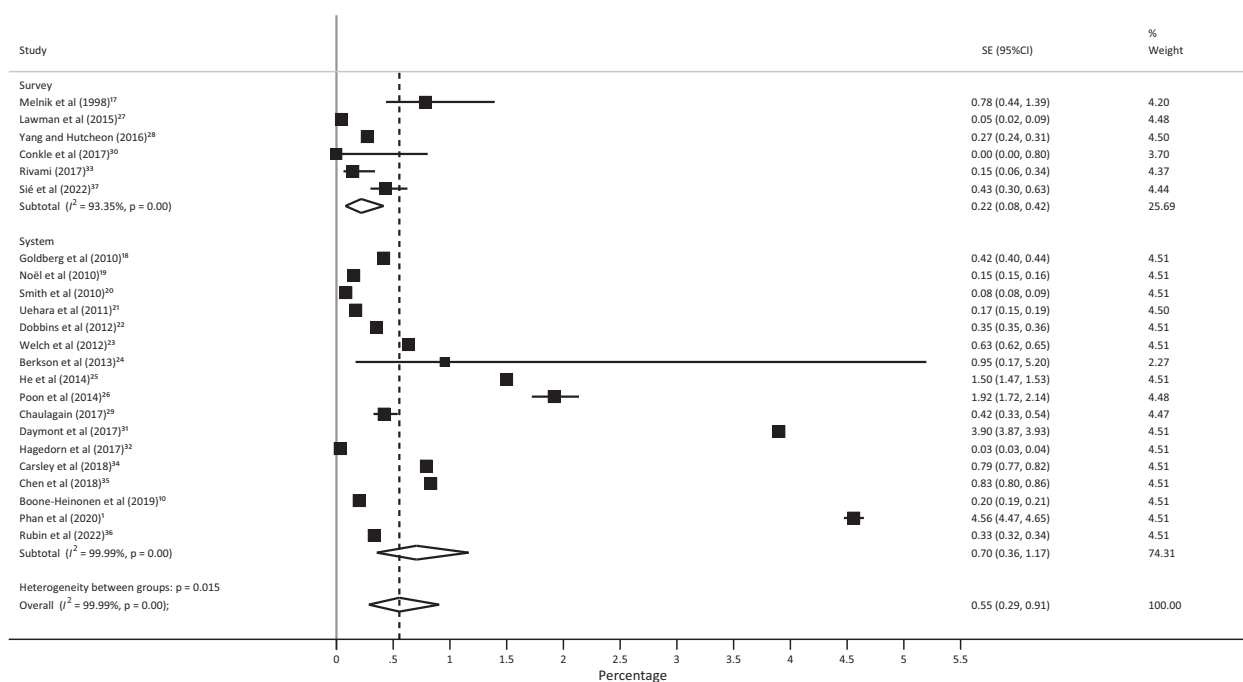


Figure 3 Frequency of implausible values for weight measure by data collection source.

designs. Third, the frequency of implausible values was greater for height than for weight.

A varied set of indicators to describe anthropometric data quality was used in the reports. In a Demographic and Health Surveys (DHS) methodological report, the following set of indicators was used to discriminate surveys: heaping of age, digit preference for weight and height, z-score SDs, and

flagged cases. The authors themselves claimed it is not sufficient to use only 1 indicator to draw conclusions about the quality of anthropometric data.⁶ Although the report is well elaborated, the task of differentiating countries' surveys with this set of indicators became difficult for the reviewers because it used multiple indicators without establishing a hierarchy among them.

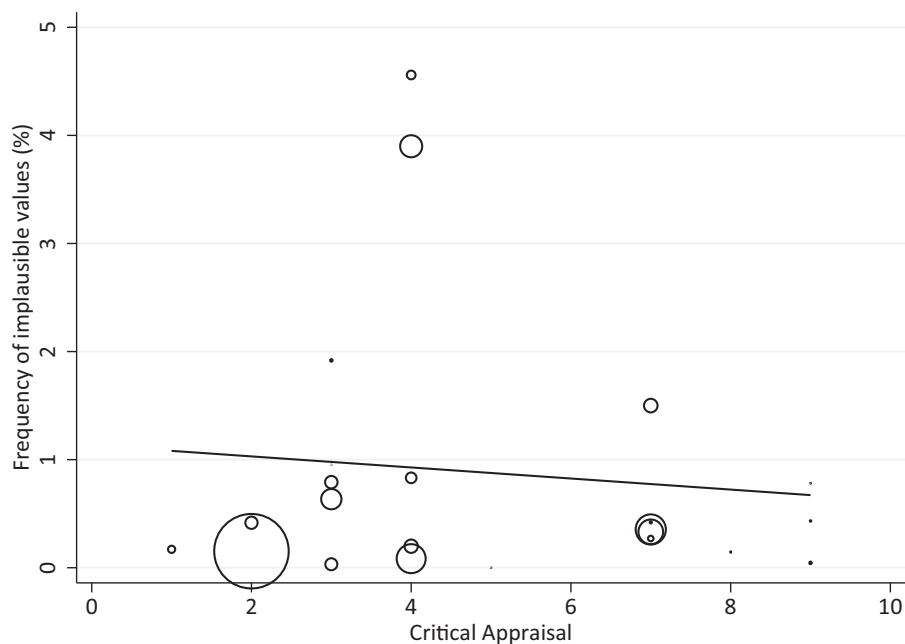


Figure 4 Frequency of implausible values for weight measure by quality score (study size is represented by the circle size).

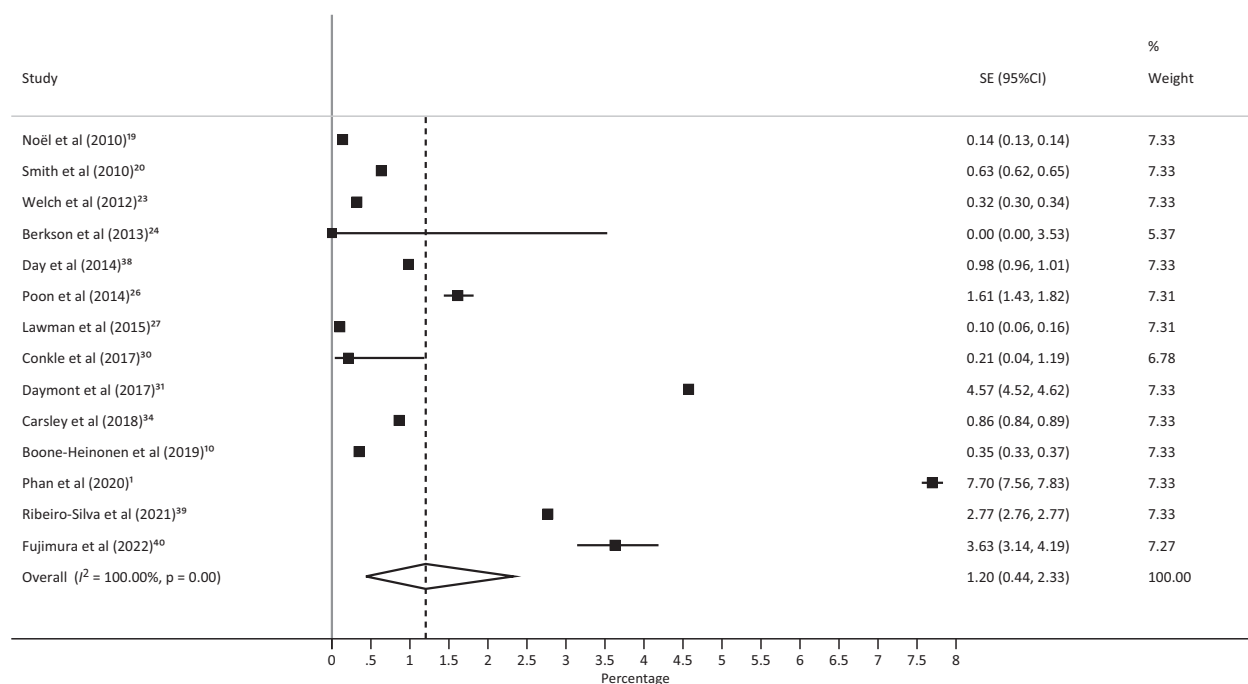


Figure 5 Frequency of implausible values for height measure.

Two approaches to summarize a group of indicators also deserve to be highlighted. The first is the method described in the SMART methodology, which gives ranges of scores for acceptability of each indicator, gives an overall score, and indicates whether the indicator is acceptable.^{41,42} The second is an approach using principal component analysis to assess anthropometric data quality to summarize a set of indicators. Perumal et al⁴³ proposed summarizing 7 indicators using data

from 145 Demographic and Health Surveys: date-of-birth completeness, completeness of anthropometric measurement, digit preference for height and age, the absolute difference in height-for-age z score by month of birth, biologically implausible values, and height-for-age and weight-for-height z scores dispersion. The study demonstrated how well a composite and multidimensional index discriminated countries' surveys in terms of excellent and poor data quality, and how

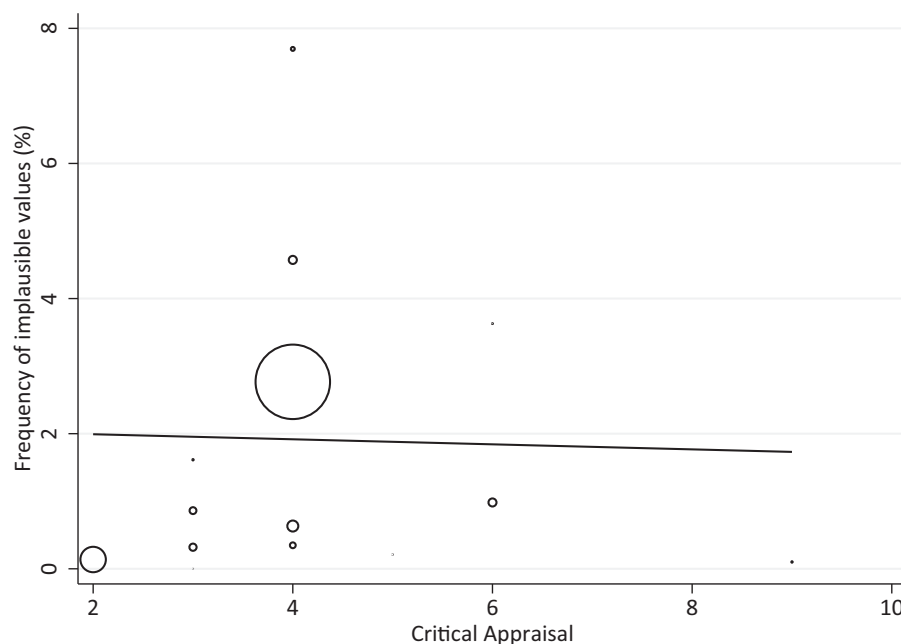


Figure 6 Frequency of implausible values for height measure by quality score (study size is represented by the circle size).

quality varied over time. This approach could be applied in other situations, such as when analyzing anthropometric data quality over time from data available in electronic health records systems.

Cross-sectional methods to determine if values are implausible mostly were used in cross-sectional and longitudinal study designs, probably due to the recommendations of institutions such as WHO, ease of application, and reproducibility. Conversely, longitudinal approaches were applicable only in cohorts and the methodologies varied due to removing lower height values than the previous one,⁴⁴ and even multilevel models.^{23,28,45} Shi et al⁴⁵ conducted a simulation in which errors in length, weight, and head circumference measurements were induced after they compared sensitivity and specificity to detect these errors among cross-sectional and longitudinal methods. They found high values of specificity (>85%) for all methods and mainly in cross-sectional approaches (>99%). On the other hand, sensitivity indicators were lower and achieved values of 10.7% to 14.1% only in longitudinal methodology.

Most methodologies to handle implausible values are based on some cutoff statistically predefined.^{46–49} For children, the references from WHO and the Centers for Disease Control and Prevention are commonly chosen in clinical practice and research.^{50,51} Freedman et al⁵² demonstrated that it is important to take into account other anthropometric variables to decide about the plausibility of a value, and they suggested modifying the cutoff given overnutrition. In their study, they compared values of weight, height, and body

mass index flagged as implausible according to WHO flags, and they concluded that a considerable number of values probably were correct. First, the authors used data from the National Health and Nutrition Examination Survey, a survey that collects anthropometric data using standardized protocols and appropriate equipment.⁵ Second, the authors could compare values of weight, height, and body mass index with other anthropometric data, namely waist circumference, arm circumference, and leg length.⁵² It would be interesting to understand how the Centers for Disease Control and Prevention cutoff classifies child's nutritional status in different contexts and using different growth references.

Another question that emerges is about SMART methodology, which is widely used in surveys in low- and middle-income countries.⁴² Attention is required when using this methodology in countries in which the double burden of malnutrition is an issue, because SMART uses the z-score mean of the sample as a reference and the cutoff is too narrow.⁴¹ For example, suppose that in a sample of children younger than 5 years, the mean of weight for height had a z score of -1 ; using SMART flags, values smaller than -4 SD or greater than 2 SD would be excluded. In that hypothetical situation, no child would be classified as overweight.

In the meta-analysis, the frequency of implausible values was higher for height than for weight, and values from 2 studies pulled the results to the right. Taking weight measurements depends basically on well-calibrated scales, and taking measurements of height depends on adequate equipment and professional

performance to carry out the procedures.⁵ For reliable height measurements, the anthropometrist has to fulfill more steps that require knowledge and precision (eg, aligning the individual's head in the Frankfort horizontal plane remains 1 of the major requirements).⁵ The frequency of implausible values for weight was lower in surveys than in electronic health records systems. In general, surveys are designed to achieve anthropometric data quality, and they follow the following set of recommendations: (1) training to accomplish a standardized measurement protocol, (2) equipment appropriated and periodically calibrated, and (3) assessment of data quality during data collection.⁵ Improvements in the anthropometric data quality available in electronic health records systems are important to promote adequate diagnosis and treatment in individual care and to monitor nutritional status in communities, hospitals, and populations. Evidence to discriminate the frequency of implausible values according to study design or approaches to sign implausible values was not found, probably because there was no difference among the stratifications or because the number of studies was too small to find significant differences.

Measurement errors might occur due to inappropriate equipment, absence of measurement procedures, and poor professional performance.⁴ The SD increases substantially as far as the proportion of measurement errors increases in the sample.⁴⁷ In distributions with high SDs, the curves tend to be broader than a distribution with SD equal to 1.² In this situation, malnutrition indicators would be overestimated because the estimates are based on the extremes of the distribution.² Therefore, excessive variability is a challenge for estimating malnutrition prevalence. In this special condition, cleaning the data with the methods found in this systematic review will not go far enough. We identified studies that adjusted the prevalence given overdispersion. Results presented by Ghosh et al⁵³ showed overdispersion in anthropometric data from National Family Survey 4 conducted in 2015–2016 in India. Using WHO flags, the unadjusted prevalence of stunting in children younger than 5 years was 36.2%, whereas the uncertainty-adjusted prevalence of stunting was 18.7%.⁵³ Likewise, Fujimura et al⁴⁰ used the PROBIT methodology to estimate the prevalence of stunting and wasting in children younger than 5 years in Namibia. The unadjusted prevalence values of wasting and stunting, using WHO flags in the Namibia Household Income and Expenditure Survey conducted in 2015–2016, were 11.2% and 30.3%, respectively, whereas the adjusted prevalence values using PROBIT for wasting and stunting were 4.2% and 20.9%, respectively.⁴⁰

This systematic review has its limitations. The inclusion criteria of study design and participants were

too large, allowing the inclusion of different types of studies and all age groups which could have caused heterogeneity. During the application of the critical appraisal, it was difficult to be sure if the procedures in the anthropometric data collection were adequate; the description of the methods reported in the studies was used as a guide. Another potential limitation was the different cutoffs used for the identification of implausible values over the growth references and the studies. The diversity of cutoffs might have implications in the estimate of implausible values and the heterogeneity observed in the results. One of the strengths of this review is the large number of reports used in the qualitative synthesis to describe the methods to analyze anthropometric data quality.

CONCLUSION

Because a multivariate and composite index summarizes a set of indicators in a single index, which allows comparisons across countries and over time, using a multivariate index is a good option to evaluate anthropometric data quality. Multivariate techniques are extensively used in other food and nutrition areas, and the application of this methodology to evaluate anthropometric data quality is promising. The frequency of implausible values for weight was less than 1%, so it is possible to claim that weight measurements are appropriate. The frequency of implausible values for height was 1.20%, a finding that deserves attention in future studies. Researchers must be able to guarantee standardized procedures and appropriate equipment to collect anthropometric variables. Data from electronic health records are important to provide continuous training to the health workers and adequate investment in equipment and infrastructure.

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Declaration of interest. The authors have no relevant interests to declare.

Supporting Information

The following Supporting Information is available through the online version of this article at the publisher's website.

[Appendix S1](#) Terms used in the databases searches

[Appendix S2](#) Critical appraisal form

[Appendix S3](#) List of reports of the included studies (N = 123)

[Appendix S4](#) Reason for exclusion of reports assessed for data extraction (n = 8)

[Table S1](#) Descriptive data for all 123 reports of included studies

[Table S2](#) Critical appraisal for all 123 reports of included studies

[Figure S1](#) Frequency of implausible values for weight measure by study design

[Figure S2](#) Frequency of implausible values for weight measure by identification method

[Figure S3](#) Frequency of implausible values for weight measure by life cycle

[Figure S4](#) Frequency of implausible values for weight measure in each study distributed according to the standard error

[Figure S5](#) Frequency of implausible values for height measure by study design

[Figure S6](#) Frequency of implausible values for height measure by life cycle

[Figure S7](#) Frequency of implausible values for height measure in each study distributed according to the standard error

Data Availability

Data are available upon request to the authors.

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